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Introduction To Algorithms

Week 6 - Graph Traversal and Modeling

"Modeling Connected Things"

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Terms to Remember

Term	General Meaning	Graph / Network Meaning
Network	Any connected system	A graph model of connected nodes and edges
Node	A point or object	A specific item in the graph
Vertex	A corner or point	Another term for node
Edge	Outside boundary	A relationship or connection between nodes
Link	A connection	Another term for edge
Neighbor	Something nearby	A node directly connected to another node
Path	A route	A sequence of connected nodes
Traversal	Moving through something	Visiting graph nodes using an algorithm
Centrality	Importance	A metric estimating structural importance
Density	How compact something is	How many possible connections actually exist

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```
... mirror_mod = modifier_ob...
... mirror_mod.mirror_object =
operation == "MIRROR_X":
    mirror_mod.use_x = True
    mirror_mod.use_y = False
    operation == "MIRROR_Y":
    mirror_mod.use_x = False
    mirror_mod.use_y = True
    operation == "MIRROR_Z":
    mirror_mod.use_x = False
    mirror_mod.use_y = False
    mirror_mod.use_z = True
selection at the end: add
obj.select-1
... instance.objects.action
["Selected" + str(modifier
mirror_ob.select-0
= bpy.context.selected_obj
data.objects[one.name].sel
int("please select exactly
... OPERATOR CLASSES ...
types.Operator):
    X mirror to the selecte
object.mirror_mirror_x"
mirror_x"
context):
    next.active_object is m
```

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What We Will Use Today

- ▶ Graph vocabulary
- ▶ Directed, Undirected, Simple, and Weighted graph recognition
- ▶ Adjacency-list representation
- ▶ Neighborhoods and shortest-path intuition
- ▶ BFS and DFS traversal
- ▶ Queue and Stack state as evidence
- ▶ Model-limit explanation

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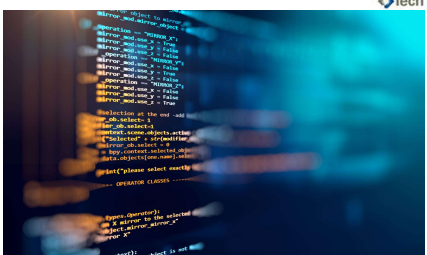
What We Will “Touch On” and Revisit Later

- ▶ similarity and recommendation
- ▶ ranking and centrality ideas
- ▶ AI/data relationship modeling
- ▶ explainability and limits of graph-based claims
- ▶ optional social network analysis context



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Review Coding Lab

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Review: What Lab 05 Taught Us

Last week - strategy choice depended on problem shape:

- correctness
- readability
- growth
- fit to data
- limitations

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Today's Question

How can we model connected parts so an algorithm can move through them?

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Graphs Model Relationships

A graph is a model of connected things.

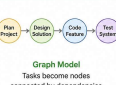
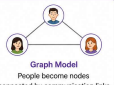
It can represent:

- ▶ places connected by routes
- ▶ tasks connected by dependencies
- ▶ people connected by communication
- ▶ steps connected in a workflow
- ▶ rooms connected by paths

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From Real-World Things to Graphs

1 Buildings	2 Tasks	3 People
 Real World	 Real World	 Real World
 Graph Model Buildings become nodes connected by walking paths.	 Graph Model Tasks become nodes connected by dependencies.	 Graph Model People become nodes connected by communication links.

Graphs model relationships.

Graph Model

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What Counts as Success Today

- define nodes and edges
- build an adjacency list
- trace BFS or DFS
- explain traversal evidence
- name what the graph le

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Textbook Review

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Graph Algorithm Overview

The reading covers:

1. graph representation
2. network theory analysis
3. graph traversals
4. case study
5. neighborhood techniques

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Terms Become Precise

Some familiar words become technical:

- ▶ network
- ▶ node
- ▶ edge
- ▶ neighbor
- ▶ path
- ▶ traversal
- ▶ density
- ▶ Centrality

Use the graph meaning during this lesson

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Everyday Meaning	Graph Meaning
 Network = connected people or systems	 Network = nodes connected by edges
 Neighbor = someone nearby	 Neighbor = a directly connected node

 Words change meaning in a graph context.

Term Precision

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Textbook Review

Do Not Panic Over Formulas

The reading includes advanced graph formulas.

For this week:

- ▶ notice the term
- ▶ read for the idea
- ▶ connect it to a simple graph
- ▶ do not try to master every formula

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Advanced Reference

$$\sum_{v \in V} \deg(v) = 2|E|$$

$$|V| - |E| + c = k$$

$$A^T = A$$

$$d(v) = |N(v)|$$

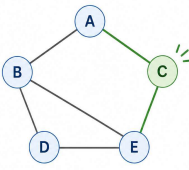
$$\sum_{u \in E} w(u)$$

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$$



Use later when you need them.

Simple Graph (5 nodes, 6 edges)



 **Start with the model: nodes and edges.**
 Build intuition first. Formulas are references.

Formula Posture

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Graph Basics

Core graph pieces:

- ▶ vertices or nodes
- ▶ edges or links
- ▶ network structure
- ▶ simple, directed, undirected, weighted types

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Algorithm

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Neighborhoods

A neighborhood asks:

- ▶ What is directly connected?
- ▶ What is one hop away?
- ▶ What is two hops away?
- ▶ What can be reached from here?

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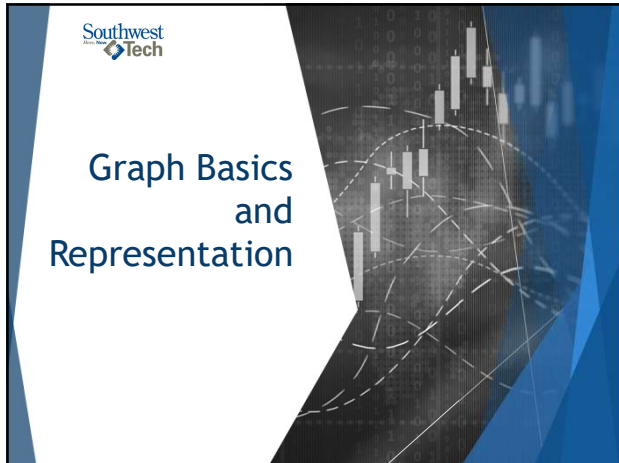
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Traversals

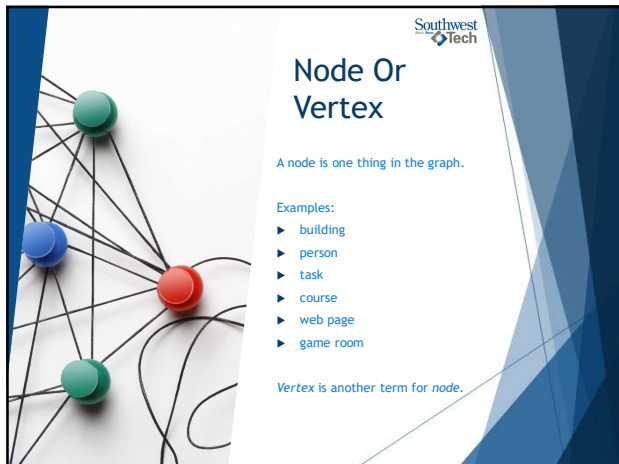
The reading introduces:

- ▶ breadth-first search (BFS)
- ▶ depth-first search (DFS)
- ▶ traversal order
- ▶ specific searches using traversal

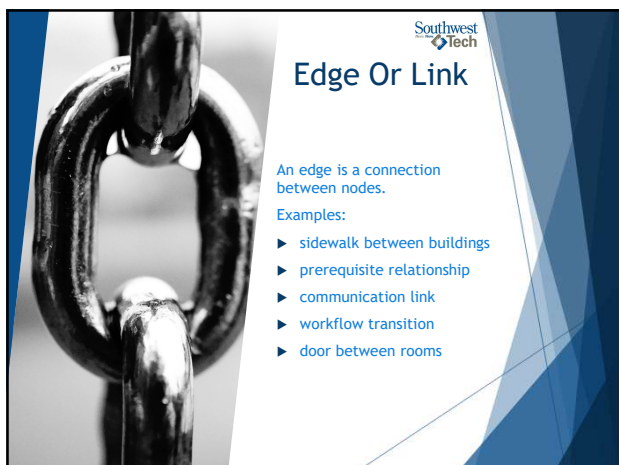
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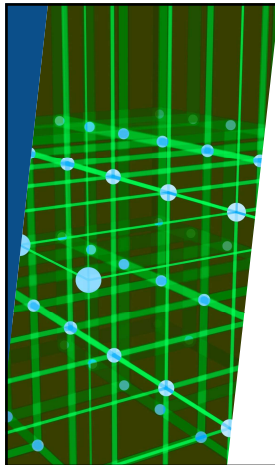
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Network

In graph work, a network means:

- ▶ defined nodes
- ▶ defined edges
- ▶ a model of relationships

It does not mean "everything about the real system."

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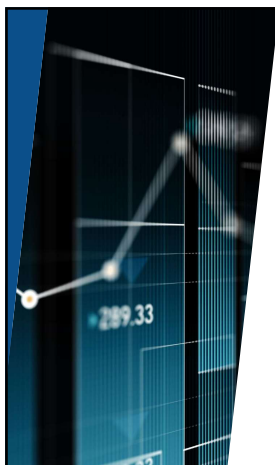
Simple Graph

A simple graph usually avoids:

- ▶ duplicate edges
- ▶ self-loops
- ▶ extra edge complexity

It is often a good beginner model.

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Directed Graph

A directed graph has one-way relationships.

Examples:

- ▶ course prerequisite
- ▶ workflow step
- ▶ escalation path
- ▶ one-way route

Direction changes what can be reached.

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Undirected Graph

An undirected graph has two-way relationships.

Examples:

- ▶ sidewalk between buildings
- ▶ hallway between rooms
- ▶ mutual connection
- ▶ two-way transit route

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Weighted Graph

A weighted graph adds a value to edges.

Weights may represent:

- ▶ distance
- ▶ time
- ▶ cost
- ▶ risk
- ▶ strength of connection

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Adjacency List

An adjacency list stores each node with its neighbors.

Example:

Parking: Library, Student Center
 Library: Parking, Science Hall, Cafeteria

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Graph

Adjacency List

A	B, C
B	A, D
C	A, D
D	B, C, E
E	D

💡 Each list entry names direct neighbors.

Adjacency List

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Diagram And Adjacency List Must Agree

Check the model both ways:

- ▶ Does every diagram connection appear in the adjacency list?
- ▶ Does every adjacency-list connection appear in the diagram?
- ▶ Are directed edges shown clearly?
- ▶ Are two-way edges listed both ways?

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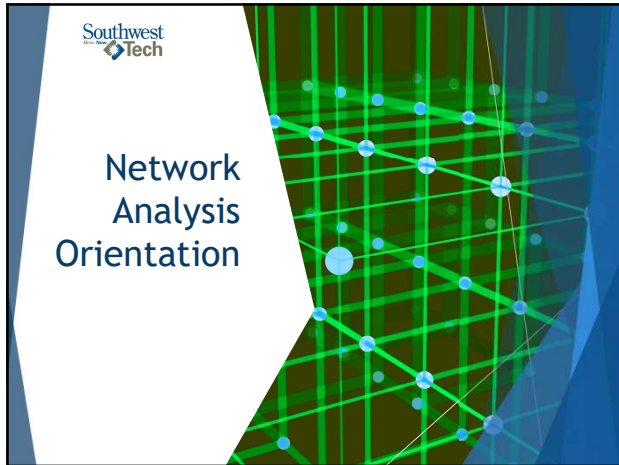
A Graph is a Model

A graph may leave out:

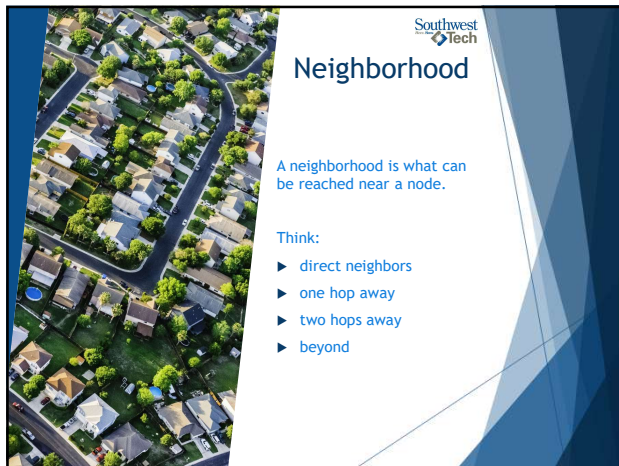
- ▶ distance
- ▶ time
- ▶ priority
- ▶ capacity
- ▶ direction
- ▶ human judgment
- ▶ real-world messiness

A model is useful, but incomplete.

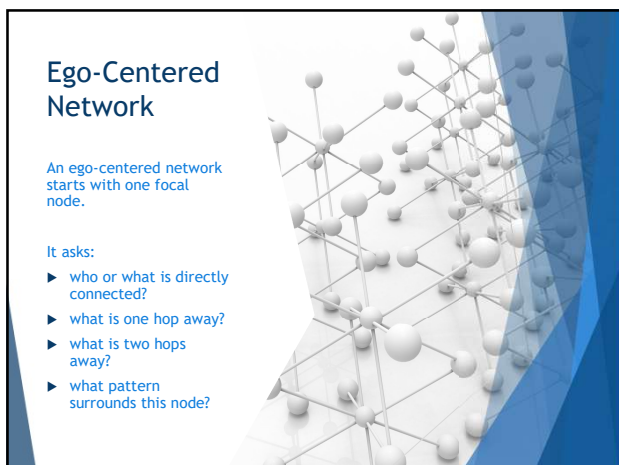
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Shortest Path

Shortest path asks:

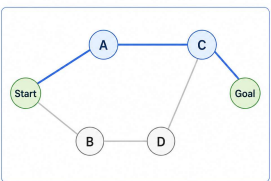
- ▶ what route uses the fewest steps?
- ▶ what route has the lowest cost?
- ▶ what route reaches the target most directly?

The answer depends on the model.

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Shortest Path in an Unweighted Graph



Route 1:
Start → A → C → Goal
Fewest edges

Route 2:
Start → B → D → C → Goal
Longer route

 In an unweighted graph, shortest usually means fewest edges.

Shortest Path

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Creating A Neighborhood

To create a neighborhood:

1. choose a start node
2. find direct neighbors
3. find neighbors of neighbors
4. decide how far to expand
5. state what the neighborhood means



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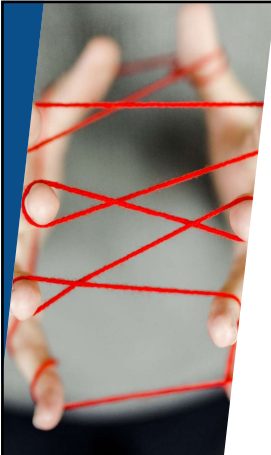
**Optional Metrics:
Recognition Only**

The optional reading includes:

- ▶ triangle
- ▶ density
- ▶ degree
- ▶ betweenness
- ▶ closeness
- ▶ eigenvector centrality

For this course: recognize the ideas.

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Degree And Density

Degree:

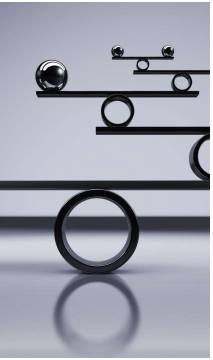
- ▶ How many edges connect to a node?

Density:

- ▶ How many possible connections actually exist?

Both describe structure.

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Centrality Is Not Importance By Itself

Centrality measures estimate structural position.

They do not automatically prove:

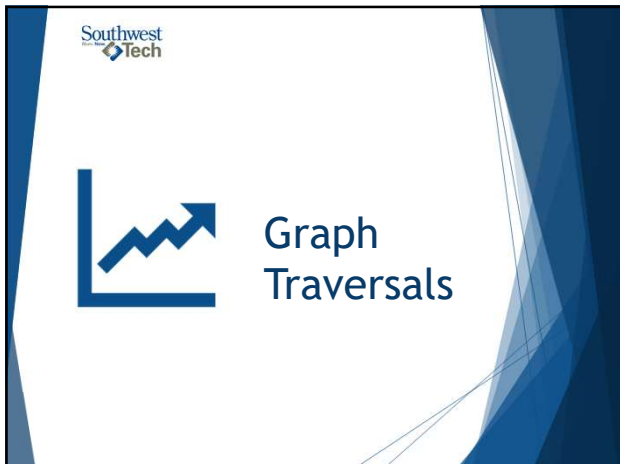
- ▶ value
- ▶ authority
- ▶ quality
- ▶ fairness
- ▶ truth

Metrics need interpretation.

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Breadth-First Search

BFS explores outward by layers.

It tends to visit:

- ▶ the start node
- ▶ direct neighbors
- ▶ neighbors of neighbors
- ▶ later layers

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Step	Current Node	QUEUE State	Visited Nodes
1			
2 ...			

BFS Evidence

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BFS Layers from Start Node A

Queue Evidence

Step	Visit	Queue
1	A	B, C
2	B	C, D
3	C	D, E
4	D	E, F

BFS explores nearest neighbors first.

BFS Evidence

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Depth-First Search

DFS follows a path deeply before backing up.

It tends to:

- ▶ choose a neighbor
- ▶ continue from that neighbor
- ▶ keep going until it cannot
- ▶ backtrack and try another path

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Step	Current Node	STACK State	Visited Nodes
1			
2 ...			

DFS Evidence

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DFS Deep Dive (Current Path)

Following:
A → B → D → F

Stack Evidence

Step	Visit	Stack
1	A	C, B
2	B	C, D
3	D	C, F
4	F	C

DFS follows one path before backing up.

DFS Evidence

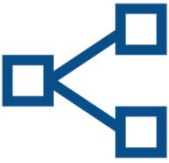
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Question	BFS	DFS
Explores how?	outward by layers	deeply by path
Common state	queue	stack or call stack
Useful for	near/reachable layers	path exploration

BFS vs DFS

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Neighbor Order Matters

Traversal order can change when:

- ▶ the start node changes
- ▶ the neighbor order changes
- ▶ the graph is directed
- ▶ the traversal method changes

The evidence should name the conditions.

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Optional Network Context



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Visualizing A Graph

A graph can be shown as:

- ▶ diagram
- ▶ adjacency list
- ▶ traversal table
- ▶ Python output
- ▶ library visualization

Each view reveals something different.

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Social Network Analysis Context

Social network analysis uses graphs to study relationships.

It may ask:

- ▶ who is connected?
- ▶ who bridges groups?
- ▶ where are clusters?
- ▶ how dense is the network?

Use caution when interpreting results.

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Fairness And Closeness

Graph metrics can influence decisions.

Ask:

- ▶ What data created the graph?
- ▶ What relationships were included?
- ▶ What relationships were ignored?
- ▶ Who could be misrepresented?

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What To Deeply Learn vs Recognize



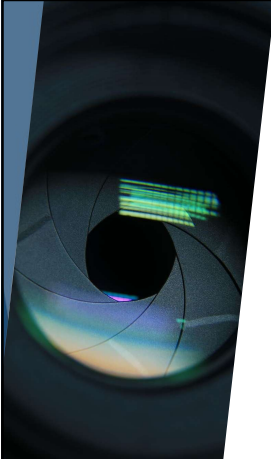
Deeply learn now:

- ▶ nodes and edges
- ▶ adjacency list
- ▶ BFS and DFS
- ▶ traversal evidence
- ▶ model limits

Recognize for later:

- ▶ centrality metrics
- ▶ SNA
- ▶ graph libraries

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Demo

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Demo Scenario

Problem:

Model a small campus route map with buildings as nodes and sidewalks as edges

Goal

Demonstrate:

- a text diagram of the route map
- adjacency-list creation
- BFS traversal from one start building
- DFS traversal from the same start building

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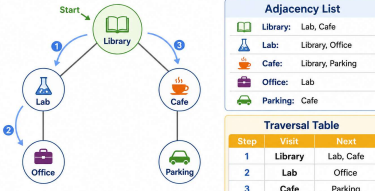
Demo Evidence

- ▶ text diagram
- ▶ adjacency list
- ▶ BFS traversal table
- ▶ DFS traversal table
- ▶ traversal summary

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Campus Graph Model with Traversal Evidence



Adjacency List

- Library: Lab, Cafe
- Lab: Library, Office
- Cafe: Library, Parking
- Office: Lab
- Parking: Cafe

Traversal Table

Step	Visit	Next
1	Library	Lab, Cafe
2	Lab	Office
3	Cafe	Parking

 We model places as nodes and paths as edges.
Traversals help us explore the campus.

Demo Evidence

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Demo Key Questions



- ▶ What is the current node?
- ▶ What neighbors are available?
- ▶ What is waiting in the queue or stack?
- ▶ What has already been visited?
- ▶ What does the model leave out?

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Demo Explanation

"The graph model stayed the same, but BFS and DFS visited nodes in different orders because BFS explores outward by layers while DFS follows a path more deeply before backing up"

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From Demo to Lab

In Lab 06, you will model a different system.

Options include:

- ▶ workflow steps
- ▶ escalation paths
- ▶ transit stops
- ▶ game rooms
- ▶ prerequisites
- ▶ communication network
- ▶ grid movement map

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Lab 06 Evidence

at least 6 nodes	at least 7 edges	adjacency list
diagram or text representation	BFS or DFS traversal	traversal table
model limitation		

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README Evidence



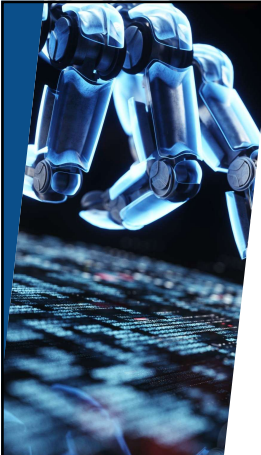
Your README should show:

- ▶ What the graph models
- ▶ What the nodes represent
- ▶ What the edges represent
- ▶ What traversal you used
- ▶ What the traversal shows
- ▶ What the graph leaves out
- ▶ AI-use note, if applicable

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AI Use In Lab 0n



Manual first:

- ▶ draft your graph
- ▶ write your adjacency list
- ▶ choose your start node
- ▶ attempt traversal evidence

AI-assisted after:

- ▶ check diagram/list consistency
- ▶ explain BFS or DFS vocabulary
- ▶ critique model limits

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Wrap up



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What To Carry Forward

Graphs help us reason about connected systems.

Carry forward:

- ▶ Model the relationships
- ▶ Make connections explicit
- ▶ Trace the traversal
- ▶ Explain what the model shows
- ▶ State what the model leaves out

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Lab On Success Check

- ▶ Represents relationships clearly
- ▶ Uses graph vocabulary correctly
- ▶ Includes adjacency-list evidence
- ▶ Shows BFS or DFS behavior
- ▶ Explains model limits

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Next Step

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For Lab 06:

- choose a system
- identify nodes and edges
- build an adjacency list
- create a diagram or text representation
- run or simulate BFS or DFS
- explain what the model leaves out

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How To Use The Textbook For Next Week's Reading

Next week connects algorithms to AI, data, recommendations, and security.

As you read:

- ▶ Focus on representation before advanced methods
- ▶ Notice how similarity and recommendation depend on chosen features
- ▶ Skim advanced ML language for recognition
- ▶ Learn the security vocabulary without trying to master cryptography
- ▶ Watch for limitations such as cold start, sparse data, and weakest link



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Thank you!

Have Fun!

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